KAIZEN MATHS

Lesson starter

Friction and Rough Surfaces

Rough Inclined Planes

Name _____

S1

A block is just on the point of sliding down a rough plane inclined at 30° to the horizontal. Find the coefficient of friction.

S2

A block of mass 10 kg slides down a rough plane inclined at 35°. The coefficient of friction is 0.15. Find its acceleration. Take $g = 9.8\ m\ s^{-2}$.

S3

A block of mass 8 kg is pulled up a rough plane inclined at 15° at constant speed. The coefficient of friction is 0.2. Find the pulling force parallel to the plane. Take $g = 9.8\ m\ s^{-2}$.

S4

A block rests on a rough plane. The coefficient of friction is 0.6. Find the greatest angle the plane can make with the horizontal before the block begins to slide.

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MEC-DED4-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Inclined Planes

Name _____

EA1

A block is just on the point of sliding down a rough plane inclined at 25° to the horizontal. Find the coefficient of friction.

EA2

A block of mass 15 kg slides down a rough plane inclined at 40°. The coefficient of friction is 0.1. Find its acceleration. Take $g = 9.8$ m s^{-2}.

Show your working

KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Inclined Planes

Name _____

EB1

A block of mass 5 kg is pulled up a rough plane inclined at 20° at constant speed. The coefficient of friction is 0.1. Find the pulling force parallel to the plane. Take $g = 9.8$ m s^{-2}.

Show your working

KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Inclined Planes

Name _____

EA1

A block is just on the point of sliding down a rough plane inclined at 25° to the horizontal. Find the coefficient of friction.

EA2

A block of mass 15 kg slides down a rough plane inclined at 40°. The coefficient of friction is 0.1. Find its acceleration. Take $g = 9.8$ m s^{-2}.

MEC-DED4-L3 · EA

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KAIZEN MATHS

Exit ticket

Friction and Rough Surfaces

Rough Inclined Planes

Name _____

EB1

A block of mass 5 kg is pulled up a rough plane inclined at 20° at constant speed. The coefficient of friction is 0.1. Find the pulling force parallel to the plane. Take $g = 9.8$ m s^{-2}.

MEC-DED4-L3 · EB

Show your working

Teacher answers & working

Friction and Rough Surfaces · Rough Inclined Planes

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S1

A block is just on the point of sliding down a rough plane inclined at 30° to the horizontal. Find the coefficient of friction.

Answer:

$\mu = 0.577$

Working:

At limiting equilibrium, friction acts up the plane and F = μR.

Component down the plane = mg sin 30°

Normal reaction = mg cos 30°

$mg \sin 30° = \mu mg \cos 30°$

$\mu = \frac{mg\sin 30°}{mg\cos 30°}$

$\mu = \tan 30°$

$\mu = 0.577$

S2

A block of mass 10 kg slides down a rough plane inclined at 35°. The coefficient of friction is 0.15. Find its acceleration. Take g = 9.8 m s^{-2}.

Answer:

4.417 m s^{-2}

Working:

Resolve perpendicular to the plane.

$R = mg \cos 35° = 10 \times 9.8 \times \cos 35° = 80.277 \text{ N}$

Find friction. It acts up the plane because the block slides down.

$F = \mu R = 0.15 \times 80.277 = 12.042 \text{ N}$

Resolve down the plane.

$mg \sin 35° - F = ma$

$56.21 - 12.042 = 10a$

$a = \frac{44.169}{10}$

$a = 4.417 \text{ m s}^{-2}$

Teacher answers & working

Friction and Rough Surfaces · Rough Inclined Planes

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S3

A block of mass 8 kg is pulled up a rough plane inclined at 15° at constant speed. The coefficient of friction is 0.2. Find the pulling force parallel to the plane. Take g = 9.8 m s^{-2}.

Answer:

35.437 N

Working:

At constant speed, the resultant force along the plane is zero.

$R = mg \cos 15° = 75.729$ N

Friction = μR = 0.2 × 75.729 = 15.146 N

Since motion is up the plane, friction acts down the plane.

Component of weight down plane = mg sin 15° = 20.291 N

$P = mg \sin 15° + F$

$P = 20.291 + 15.146$

$P = 35.437$ N

S4

A block rests on a rough plane. The coefficient of friction is 0.6. Find the greatest angle the plane can make with the horizontal before the block begins to slide.

Answer:

31°

Working:

At the greatest angle, the block is in limiting equilibrium and is about to slide down.

$mg \sin \theta = \mu mg \cos \theta$

$\tan \theta = \mu$

$\tan \theta = 0.6$

$\theta = \tan^{-1}(0.6)$

$\theta = 31°$

Teacher answers & working

Friction and Rough Surfaces · Rough Inclined Planes

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EA1

A block is just on the point of sliding down a rough plane inclined at 25° to the horizontal. Find the coefficient of friction.

Answer:

$\mu = 0.466$

Working:

At limiting equilibrium, friction acts up the plane and F = μR.

Component down the plane = mg sin 25°

Normal reaction = mg cos 25°

$mg \sin 25° = \mu mg \cos 25°$

$\mu = \frac{mg\sin 25°}{mg\cos 25°}$

$\mu = \tan 25°$

$\mu = 0.466$

EA2

A block of mass 15 kg slides down a rough plane inclined at 40°. The coefficient of friction is 0.1. Find its acceleration. Take g = 9.8 m s^{-2}.

Answer:

5.549 m s^{-2}

Working:

Resolve perpendicular to the plane.

$R = mg \cos 40° = 15 \times 9.8 \times \cos 40° = 112.609 \text{ N}$

Find friction. It acts up the plane because the block slides down.

$F = \mu R = 0.1 \times 112.609 = 11.261 \text{ N}$

Resolve down the plane.

$mg \sin 40° - F = ma$

$94.49 - 11.261 = 15a$

$a = \frac{83.229}{15}$

$a = 5.549 \text{ m s}^{-2}$

Teacher answers & working

Friction and Rough Surfaces · Rough Inclined Planes

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EB1

A block of mass 5 kg is pulled up a rough plane inclined at 20° at constant speed. The coefficient of friction is 0.1. Find the pulling force parallel to the plane. Take g = 9.8 m s^{-2}.

Answer:

21.363 N

Working:

At constant speed, the resultant force along the plane is zero.

$R = mg \cos 20° = 46.045$ N

Friction = μR = 0.1 × 46.045 = 4.604 N

Since motion is up the plane, friction acts down the plane.

Component of weight down plane = mg sin 20° = 16.759 N

$P = mg \sin 20° + F$

$P = 16.759 + 4.604$

$P = 21.363$ N